

Expt 4. Design of Low Pass Filter using Windowing Techniques

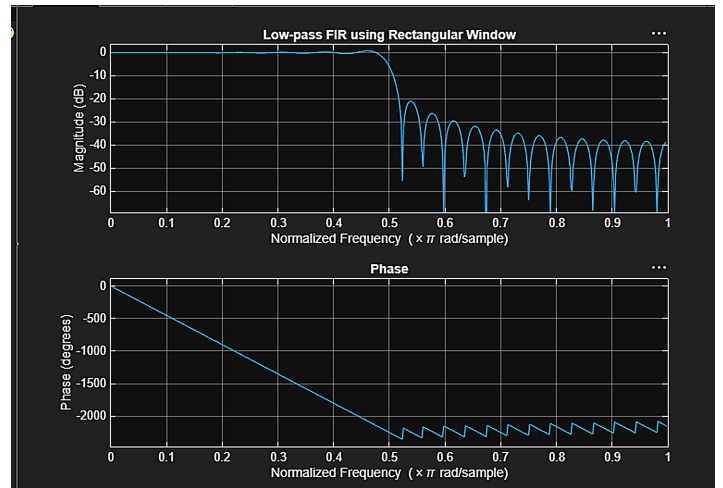
4.1 Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique

```
1 clc;
2 clear;
3 close all;
4 % --- Input Parameters ---
5 fp = 1000; % Passband cutoff (Hz)
6 fs = 4000; % Sampling frequency (Hz)
7 N = 50; % Filter order (even number preferred)
8 wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
9 % --- 1. FIR Low Pass using Rectangular Window ---
10 h_rect = fir1(N, wc, 'low', rectwin(N+1));
11 disp('--- Rectangular Window Coefficients ---');
12 disp(h_rect);
13 % --- 2. FIR Low Pass using Hamming Window ---
14 h_ham = fir1(N, wc, 'low', hamming(N+1));
15 disp('--- Hamming Window Coefficients ---');
16 disp(h_ham);
17 % --- 3. FIR Low Pass using Hanning Window ---
18 h_han = fir1(N, wc, 'low', hanning(N+1));
19 disp('--- Hanning Window Coefficients ---');
20 disp(h_han);
21 % --- Plot Frequency Responses ---
22 figure;
23 freqz(h_rect,1);
24 title('Low-pass FIR using Rectangular Window');
25 figure;
26 freqz(h_ham,1);
27 title('Low-pass FIR using Hamming Window');
28 figure;
29 freqz(h_han,1);
30 title('Low-pass FIR using Hanning Window');
```

Command Window

New to MATLAB? See resources for [Getting Started](#)

```
--- Rectangular Window Coefficients ---
0.0126
0
-0.0137
0
0.0159
0
-0.0166
0
0.0185
0
-0.0219
0
0.0242
```



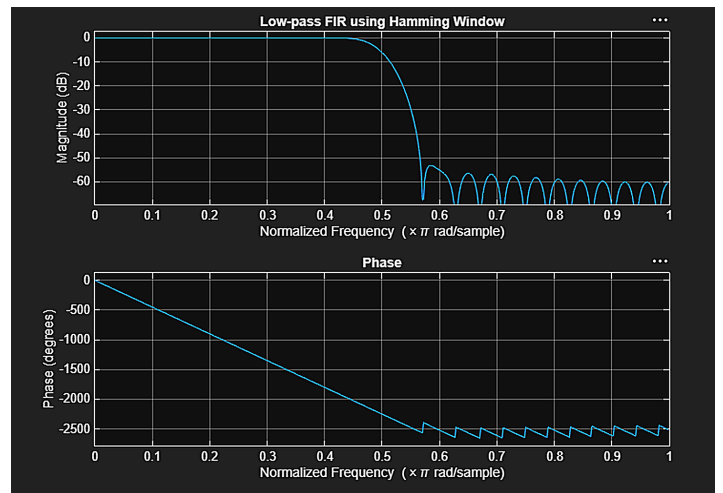
4.2 Design and Analysis of an FIR Low-Pass Filter Using the Hamming Window Technique

```
1 clc;
2 clear;
3 close all;
4 % --- Input Parameters ---
5 fp = 1000; % Passband cutoff (Hz)
6 fs = 4000; % Sampling frequency (Hz)
7 N = 50; % Filter order (even number preferred)
8 wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
9 % --- FIR Low Pass using Hamming Window ---
10 h_ham = fir1(N, wc, 'low', hamming(N+1));
11 disp('--- Hamming Window Coefficients ---');
12 disp(h_ham);
13 % --- Plot Frequency Responses ---
14 figure;
15 freqz(h_ham,1);
16 title('Low-pass FIR using Hamming Window');
```

Command Window

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```
-0.0084
0
0.0055
0
-0.0034
0
0.0021
0
-0.0013
0
-0.0010
0.0010
```



4.3 Design and Analysis of an FIR Low-Pass Filter Using the Hanning Window Technique

```
1 clc;
2 clear;
3 close all;
4 % --- Input Parameters ---
5 fp = 1000; % Passband cutoff (Hz)
6 fs = 4000; % Sampling frequency (Hz)
7 N = 50; % Filter order (even number preferred)
8 wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)
9 % --- FIR Low Pass using Hanning Window ---
10 h_han = fir1(N, wc, 'low', hanning(N+1));
11 disp('--- Hanning Window Coefficients ---');
12 disp(h_han);
13 % --- Plot Frequency Responses ---
14 figure;
15 freqz(h_han,1);
16 title('Low-pass FIR using Hanning Window');
```

Command Window

New to MATLAB? See resources for [Getting Started](#)

```
-0.0081
0
0.0058
0
-0.0028
0
0.0013
0
-0.0008
0
0.0000
```

>>

